

- Volatility measures (rolling standard deviation)
 - Moving averages (short and long term)
 - Volume analysis (relative volume compared to average)
5. Create a function to resample the data to different timeframes:
- Convert daily data to weekly and monthly views
 - Ensure OHLCV data is properly aggregated during resampling
6. Generate a comprehensive report on the data quality:
- Percentage of missing values found and remediated
 - Number of outliers detected
 - Correlation matrix between different assets
 - Basic statistical summary of each asset

This project will give you hands-on experience with the entire financial data pipeline - from acquisition to cleaning to transformation - preparing you for the analysis and strategy development phases of algorithmic trading.

Project 10: Market Event Impact Analyzer

Objective: Build a tool that analyzes how specific market events affect financial assets and identifies patterns that could be exploited in trading strategies.